

For Data Analytics:**1. Which functions/algorithms you have completed in the Method you designed last week? Using with software?**

I completed the Regression Analysis (log) and K-Fold Cross Validation (k=5) for my group last week using R.

I have not received the Random Forest results yet from my groupmate.

2. Please share your results, findings, and thoughts:

- **Any "expected" results? List the expected results and state why they are expected.**
 - log_avgPlaytime and sqrt_price are expected to be strong predictors of revenue.
- **Any "surprising" results? List the surprising results and state why you are surprised.**
 - The overall $R^2 = 0.313$ meaning the model explains only 31.3% of the variance in log revenue. I expected it to be lower due to the limitations, but this suggests revenue variance is driven by factors not captured in the dataset.
- **Any action plans in mind at this moment?**
 - a. Add age_of_game removal to the scopes & limitations section of report.
 - b. Start writing the Regression Analysis section of the report
 - c. Waiting on the Random Forest results...

3. Attach your modelling work up to date, including but limited to print screens, R/Python files, draft report,...

Posted at the end of document

For Business Intelligence:**1. Which tools you used to design/implement your Dashboard? For example., JIRA**

The BI Dashboard has not been built yet, it will be built this week.

2. Include UML Diagrams: e.g., data flow, ER diagrams, Import/Export, Testing

- Use Case Diagrams

USE CASES

- View revenue dashboard
- Compare price tiers
- View feature importance report
 - Read insight brief
 - Filter by publisher class

ACCESS

Game Developer → Dashboard, Price tiers, Feature report

Publisher → Insight brief, Publisher filter

Data Analyst → All use cases

- Data Flow Diagram



- Entity-Relationship Diagram

PUBLISHER	GAME	PRICE TIER
 publisherId (PK)	 steamId (PK)	 tierId (PK)
publisherName	name	tierLabel
publisherClass	releaseDate	priceMin
(AAA / AA / Indie)	price	priceMax
1 → M publishes	← TARGET 1 belongs to	
	avgPlaytime	
	reviewScore	
	publisherId (FK)	
	tierId (FK)	

- Notes:
- steamId is the primary key uniquely identifying each game record
 - publisherClass (AAA / AA / Indie) is derived from the publishers column, used as Dimension 1
 - price_tier is engineered from price, used as Dimension 2 (F2P \$0 / under \$20 / \$20–\$40 / \$40+)
 - revenue is the target variable (log10 transformed); copiesSold excluded due to circularity (revenue = copiesSold × price)
 - age_of_game was derived from releaseDate but excluded from the final model, all 1,499 records carry a 2024 release date, producing near-zero variance

3. What went well? What didn't go well? What could you have done differently?

Went well:

- The regression model ran cleanly with no missing value issues after removing the Hobbyist game.
- K-Fold results were nearly identical to the full model results.
- `age_of_game` was confirmed to be removed, adding confidence to the modeling choice.

Didn't go well:

- R^2 is lower than expected, suggesting the available features do not fully capture the complexity of what drives revenue among top Steam titles.

Done differently:

- Including additional variables such as genre, DLC count, or number of supported languages could have improved model fit and captured more of the variance in revenue.

Full R Output:

```
> # -----
> # Data prep done by Ethan Rickey
> # -----
>
> library(moments)
>
> data <- read.csv("Steam_2024_bestRevenue_1500.csv")
> dim(data)
[1] 1500 11
> nrow(data)
[1] 1500
> colSums(is.na(data))
      name      releaseDate      copiesSold      price      revenue
      0           0           0           0           0
  avgPlaytime  reviewScore publisherClass  publishers  developers
      0           0           0           0           0
    steamId
      0
>
> data$releaseDate <- as.Date(data$releaseDate, format = "%d-%m-%Y")
>
> numeric_vars <- c("copiesSold", "price", "revenue", "avgPlaytime", "reviewScore")
> round(sapply(data[numeric_vars], skewness), 4)
copiesSold      price      revenue avgPlaytime reviewScore
    18.6009     1.5751    22.8971     7.1630     -2.0486
>
> k <- max(data$reviewScore) + 1
>
> data$log_copiesSold <- log10(data$copiesSold)      # substantially positive, no zeros
> data$log_revenue <- log10(data$revenue)           # substantially positive, no zeros
> data$log_avgPlaytime <- log10(data$avgPlaytime + 1) # substantially positive, has one zero
> data$sqrt_price <- sqrt(data$price)               # moderately positive, has zeros
```

```

> data$log_reviewScore <- log10(k - data$reviewScore)      # substantially negative -> reflect +
log10
>
> transformed_vars <- c("log_copiesSold", "log_revenue", "log_avgPlaytime", "sqrt_price",
"log_reviewScore")
> round(sapply(data[transformed_vars], skewness), 4)
  log_copiesSold    log_revenue log_avgPlaytime    sqrt_price log_reviewScore
           0.9807         1.1854         0.6753        -0.1579        -0.2056
>
>
>
> # -----
> # Regression Analysis (log)
> # DATA 485 | Capstone Group 5 | CV
> # -----
>
> library(caret)
>
> # --- publisherClass ---
> table(data$publisherClass)

      AA      AAA Hobbyist      Indie
146      52          1     1301
>
> data <- data[data$publisherClass != "Hobbyist", ]      # Remove Hobbyist
> data$publisher_AA <- ifelse(data$publisherClass == "AA", 1, 0)
> data$publisher_Indie <- ifelse(data$publisherClass == "Indie", 1, 0)
> # AAA = reference category (publisher_AA = 0 and publisher_Indie = 0)
>
> # --- Age of Game ---
> # double checking release date coverage
> summary(data$age_of_game)

   Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
0.008214 0.202601 0.355921 0.347846 0.509240 0.689938
> sum(is.na(data$age_of_game))
[1] 0

```

```
> table(format(data$releaseDate, "%Y"))

2024
1499
>

> # Regression Analysis Equation:  $\log_{10}(\text{Revenue}) = \beta_0 + \beta_1(\sqrt{\text{Price}}) + \beta_2(\log_{10}(k - \text{ReviewScore})) + \beta_3(\log_{10}(\text{AvgPlaytime} + 1)) + \beta_4(\text{Publisher\_AA}) + \beta_5(\text{Publisher\_Indie}) + \varepsilon$ 
>

> # --- Build regression dataset ---
> # age_of_game removed: all records have 2024 release dates,
> # producing near-zero variance and a non-significant coefficient (p = 0.720) -> Need to add to
limitations section
> reg_data <- na.omit(data[, c("log_revenue", "sqrt_price", "log_reviewScore",
+                             "log_avgPlaytime",
+                             "publisher_AA", "publisher_Indie")])
>

> # --- K-Fold Cross Validation (k=5) ---
> set.seed(123)
> train_control <- trainControl(method = "cv", number = 5)
>

> # --- Run Regression Model ---
> reg_model <- train(
+   log_revenue ~ .,
+   data      = reg_data,
+   method    = "lm",
+   trControl = train_control
+ )
>

> summary(reg_model$finalModel)

Call:
lm(formula = .outcome ~ ., data = dat)

Residuals:
    Min       1Q   Median       3Q      Max
-1.69345 -0.43090 -0.08399  0.35793  2.52286
```


Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.66381	0.12285	37.962	< 2e-16 ***
sqrt_price	0.08747	0.01072	8.160	7.02e-16 ***
log_reviewScore	0.04842	0.04023	1.203	0.229
log_avgPlaytime	0.77452	0.04327	17.899	< 2e-16 ***
publisher_AA	-0.16358	0.10025	-1.632	0.103
publisher_Indie	-0.61288	0.08926	-6.866	9.63e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6203 on 1493 degrees of freedom

Multiple R-squared: 0.3131, Adjusted R-squared: 0.3108

F-statistic: 136.1 on 5 and 1493 DF, p-value: < 2.2e-16

```
> print(reg_model$results)
```

	intercept	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
1	TRUE	0.6215512	0.3089644	0.4852836	0.01662454	0.06885321	0.01227325

```
>
```

```
> # --- Actual vs Predicted Plot ---
```

```
> reg_data$predicted <- predict(reg_model, reg_data)
```

```
>
```

```
> ggplot(reg_data, aes(x = predicted, y = log_revenue)) +
+   geom_point(alpha = 0.4, color = "steelblue") +
+   geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
+   labs(title = "Actual vs. Predicted Log Revenue",
+         x = "Predicted Log10 Revenue",
+         y = "Actual Log10 Revenue") +
+   theme_minimal()
```

Findings:

OUTPUT

Publisher Class	Count
AA	146
AAA	52
Hobbyist	1 (removed)
Indie	1,301

INTERPRETATION

The dataset contains four publisher classes: AAA (52 games), AA (146 games), Indie (1,301 games), and Hobbyist (1 game). The single Hobbyist game was removed because one observation is not enough to estimate a reliable regression effect. Publisher class was encoded as two binary dummy variables, publisher_AA and publisher_Indie, with AAA serving as the reference category. This means all publisher class coefficients in the regression output represent differences relative to AAA games.

OUTPUT

Regression Equation:
 $\log_{10}(\text{Revenue})$

$$= \beta_0 + \beta_1(\sqrt{\text{Price}}) + \beta_2(\log_{10}(k - \text{ReviewScore})) \\ + \beta_3(\log_{10}(\text{AvgPlaytime} + 1)) + \beta_4(\text{Publisher_AA}) \\ + \beta_5(\text{Publisher_Indie}) + \varepsilon$$

 $\log_{10}(\text{Revenue})$

$$= 4.664 + 0.087(\sqrt{\text{Price}}) + 0.048(\log_{10}(k - \text{ReviewScore})) \\ + 0.775(\log_{10}(\text{AvgPlaytime} + 1)) - 0.164(\text{Publisher_AA}) \\ - 0.613(\text{Publisher_Indie}) + \varepsilon$$

Regression Coefficients:

Variable	Estimate	Std. Error	t value	p-value	Sig.
(Intercept)	4.664	0.123	37.962	< 2e-16	***
sqrt_price	0.087	0.011	8.160	< 2e-16	***
log_reviewScore	0.048	0.040	1.203	0.229	
log_avgPlaytime	0.775	0.043	17.899	< 2e-16	***
publisher_AA	-0.164	0.100	-1.632	0.103	
publisher_Indie	-0.613	0.089	-6.866	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05

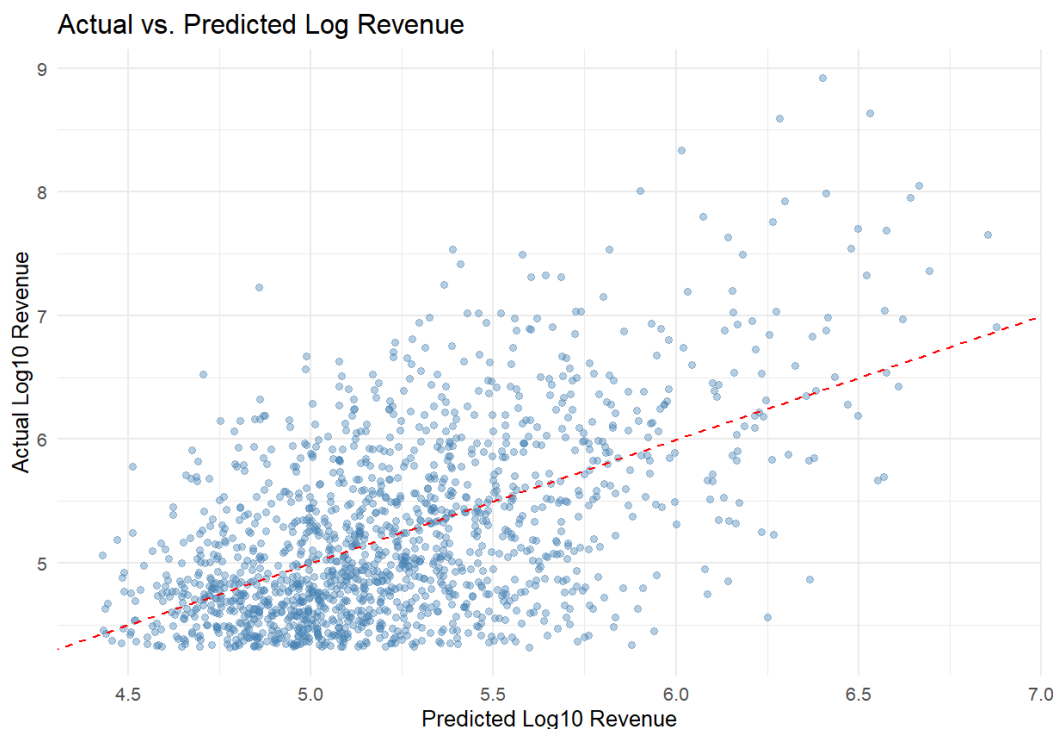
Model Performance (K-Fold Cross Validation, k=5):

R ²	Adj. R ²	RMSE (CV)	R ² (CV)
0.3131	0.3108	0.6216	0.3090

INTERPRETATION

The model explains 31.3% of the variance in log revenue. Adjusted $R^2 = 0.310$, confirming the model is not being inflated by unnecessary variables. K-Fold RMSE = 0.622 and $R^2 = 0.309$, which is nearly identical to the full model results, suggesting the model generalizes well to unseen data and is not overfitting. $p\text{-value} < 2.2e\text{-}16$, so the model as a whole is highly statistically significant.

- `log_avgPlaytime` is the strongest predictor in the model. For every one unit increase in log average playtime, log revenue increases by 0.773, holding all other factors constant. Games that players spend more time on are significantly associated with higher revenue.
 - `sqrt_price` is a significant predictor of revenue. Higher priced games tend to generate greater revenue.
 - `publisher_Indie` is associated significantly with lower revenue. Indie games earn less revenue than AAA games after controlling for all other factors.
 - `log_reviewScore` and `publisher_AA` do not significantly predict revenue. Among the top performing games, review score does not meaningfully differentiate revenue outcomes, and since AA games earn less than AAA games on average.
-

OUTPUT**INTERPRETATION**

The scatter plot shows the relationship between predicted and actual log revenue values. The red, dashed line (reference line) represents perfect prediction (predicted = actual).

Some key observations were made:

- The scatter plot shows a positive trend along the reference line.
- There are several outliers above and below the reference line, which suggests heteroscedasticity. This means the variance of errors grows for higher revenue values.
- The bulk of predictions cluster between log 4.5 and 6.0, where the model performs reasonably well, with points close to the reference line.
- There is a noticeable spread around the line which is consistent with $R^2 = 0.313$ which leaves roughly 69% of revenue variance unexplained by the available predictors.

Overall, the model demonstrates a moderate fit. It captures a general upward trend in revenue, but the prediction accuracy could be improved with additional variables.
